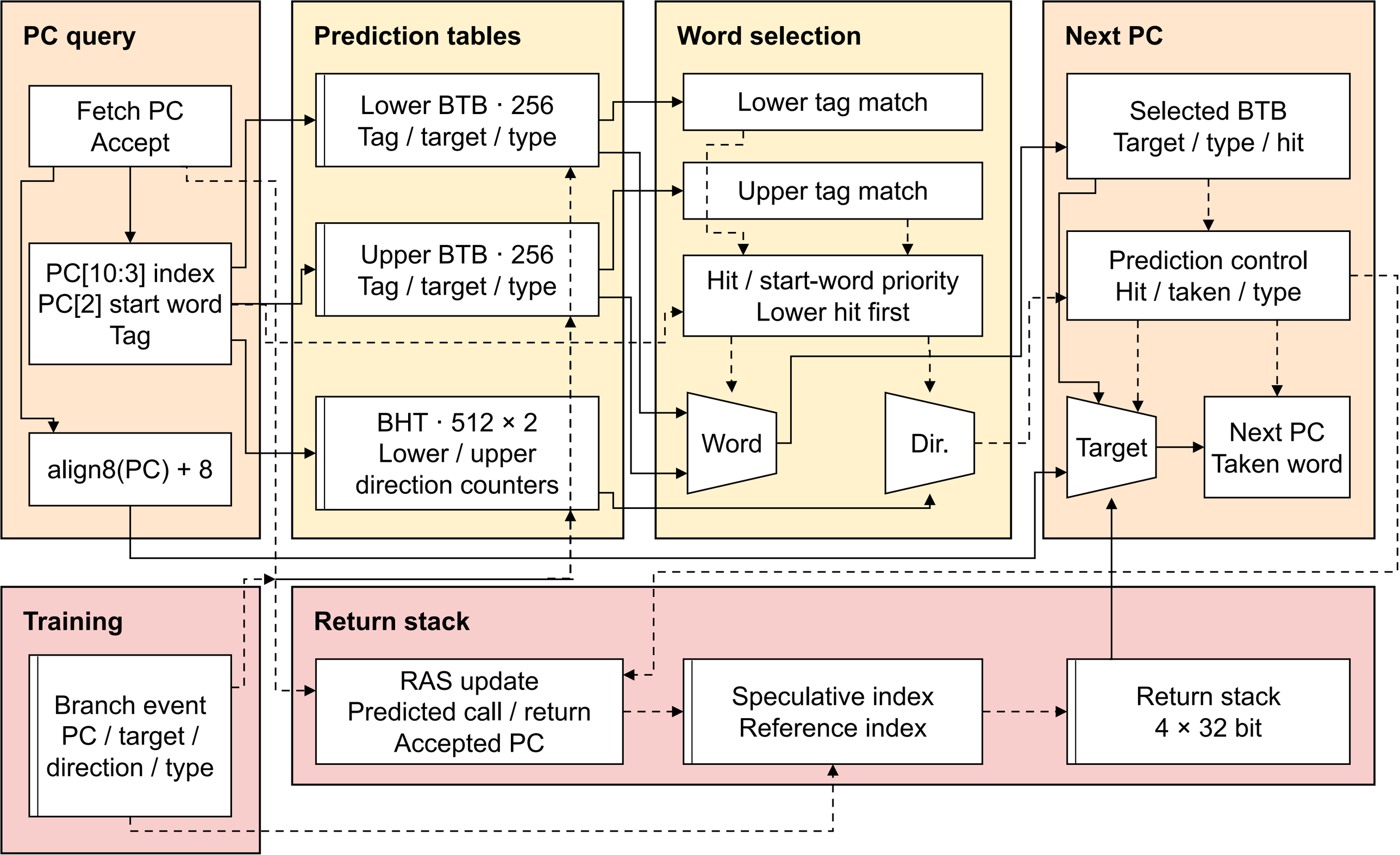
PC query
Fetch PC
Accept
PC[10:3] index
PC[2] start word
Tag
align8(PC) + 8
Prediction tables
Lower BTB · 256
Tag / target / type
Upper BTB · 256
Tag / target / type
BHT · 512 × 2
Lower / upper
direction counters
Word selection
Lower tag match
Upper tag match
Hit / start-word priority
Lower hit first
Word
Dir.
Next PC
Selected BTB
Target / type / hit
Prediction control
Hit / taken / type
Target
Next PC
Taken word
Training
Branch event
PC / target /
direction / type
Return stack
RAS update
Predicted call / return
Accepted PC
Speculative index
Reference index
Return stack
4 × 32 bit